

A GPU-Accelerated Lattice Boltzmann Solver

D2Q9 with Kokkos and MPI

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HPC with Accelerators

Roadmap: Lattice Boltzmann

- Our goal is a fluid simulation

1. Streaming

- Makes the fluid move

2. Boundary Conditions / Bounce-back & Wrap-Around

- Lets the fluid interact with its container

3. Collision

- 'Adds the physics' to the simulation

Discretization

- D2Q9 - 2 Dimensions, 9 Directions/ Channels
- Data structure: `Kokkos::View`
- Agnostic to hardware: CPU / GPU

```
typedef Kokkos::View<double***>    DISTRIB;  
typedef Kokkos::View<double**>     DENSITY;  
typedef Kokkos::View<double**[2]>   VELOCITY;
```

Algorithm: Streaming & Boundary Conditions

- Simple for loop:
 - For every value: *push* it in its direction
- Streaming directly contains wrap around logic
 - Branchless Implementation
- For Bounce back:
 - Simulate streaming on additional (ghost) layer
 - Flip afterwards

Algorithm: Collision

- For loops to calculate
 - density & average velocity
 - collision updates for distribution
- Fused into one kernel/ for loop for performance

Shear Wave Validation

- Simple initialization of epsilon using `Kokkos::sin`

Moving Lid

- For the lid:
Adjust bounce back calculations

MPI Implementation

- Divide the Boltzmann Lattice Simulation into sublattices via MPI
- Specific Implementation of MPI just for Milestone 6
- Shift over into ghost buffers (Left & Right -> Up & Down)
- Communicate ghost buffers via `MPI_Sendrecv` & `Kokkos::deep_copy`

Scaling

Logarithmic Scaling

Conclusion & Takeaways

- Impressively 'easy' to implement
 - Fluid Simulation
 - Code for GPU
- Verify your tests
- Take inspiration from other people